

The Entropy Ledger: Operational Limits for Consistent Time Travel

Spiros Ts.¹ and Collaborators²

¹*Independent Researcher*

²*Various Institutions*

We quantify retrocausal advantages in postselected time loops. A Loop Data-Processing Inequality bounds the per-attempt loop information gain by $-\log p_{\text{succ}}$ bits (plus accuracy penalties). A thermodynamic Entropy Ledger prices paradox resolution at $kT D_{\text{KL}}(P\|P^*)$. Proofs are one-shot tight and extend to multi-loop networks. Photonic/ion-trap protocols are outlined.

Model. A CPTP circuit is augmented with a heralded non-TP retro-map Π with success p_{succ} , closing a loop on success. We define monotones: temporal advantage, entropy debt.

Loop-DPI. Conditioning on success induces a Radon–Nikodym derivative bounded by $1/p_{\text{succ}}$. Thus $I(X;Y)|_{\text{succ}} \leq -\log p_{\text{succ}}$ and $\Delta I_{\odot} \leq -\log p_{\text{succ}}$ per attempt.

Entropy Ledger. Enforcing Novikov consistency projects the prior history measure onto a feasible set in KL, yielding a minimal work cost $kT D_{\text{KL}}$.

One-shot converse & multi-loop. Smooth min/max entropies imply strong converses; network cut-sets bound total advantage and include interference penalties.

Experiments. Heralded postselection in photonics and ion-traps; measured $(p_{\text{succ}}, \Delta I)$ obey the bounds.

Metaphysics (operational). Identity and free will are history-functionals constrained by the Ledger; no free chronology.
